

# Imaging the Ionosphere

Submitted by

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## **Project Overview**

The ionosphere extends from about 80 km to many hundreds kilometres in altitude. It is important for its affect on radio waves and so transmissions of electromagnetic waves by satellite-based navigation systems such as Global Positioning System (GPS) are affected as they pass through the ionosphere. GPS radio signals are slowed down as they propagate through the ionosphere, causing an increase in the propagation time of a signal when compared to the time of propagation through free space. This effect, called the propagation delay, must be taken into account when determining position using the GPS system. The ionosphere is a dispersive medium, and since the dispersion changes with frequency in a known way, a dual frequency receiver can be used to effectively eliminate this propagation delay of GPS signals caused by the ionosphere and so provide highly accurate determinations of the position of receivers monitoring GPS, or other satellite navigation systems. However, because of their relative simplicity, single frequency GPS receiver systems are often used even though they have no possibility of measuring the ionosphere delay. Instead, ionospheric models must be relied on to provide the necessary correction for the ionospheric delay. Ionospheric models generally only represent average conditions and are mostly based on ionosonde measurements that only probe the bottomside of the ionosphere, i.e. the region below the maximum electron density which generally occurs in the range 250 – 400 km altitude. Consequently it is important that the ability to model the ionosphere, and including the topside ionosphere, is improved for GPS applications, and in fact, GPS observations can themselves be used to determine properties of the ionosphere.

In fact the signal propagation delay, or position correction, is proportional to the line integral of the free electron distribution along the path of the signal from the satellite transmitter to the receiver. As explained in this thesis, this information can be used to determine the electron density distribution in the Earth's ionosphere and plasmasphere and so can be used not only to correct for the propagation delay in GPS signals, but also to study the behaviour of the ionosphere and magnetosphere. For example, changes occurring during ionospheric storms, ion composition changes, space-weather effects on telecommunications are just a few of the phenomena that can be studied.

The construction of maps of ionospheric electron density, directly from GPS measurements, is an approach to a real time monitoring of the ionosphere. During the years a variety of instruments (listed in Chapter 5 & 6) have been employed to take direct measurements of the electron density



of the ionosphere and plasmasphere. However, so far there is no universal method; each type has advantages as well as disadvantages. For example, the pulsed HF ionosonde is limited to the geometry of the reflection between the transmitter and receiver. Similarly the incoherent radar techniques are limited to the scattering point between the transmitter and receiver radars. The pulsed HF ionosonde, for instance, are not able to take measurements above the peak of the F-region.

Consequently, TEC measurements, made using transionospheric radio signals, are one of the most important methods of investigating the Earth's ionosphere and plasmasphere. However, by themselves, a series of TEC measurements are just a collection of line integrals of the free electron density and not maps of the electron density distribution or structure. In order to address the problem of imaging the electron density satisfactorily, the tomographic inversion method (based on a linear mathematical inversion) has been applied in this thesis. The tomographic inversion method essentially obtains maps of the free electron density from collections of GPS slant TECs observations obtained across a region. A prime aim of this research has been to produce such a tomographic algorithm and validate it with other independently measured data.

The tomographic inversion algorithm, which is called the Algebraic Reconstruction Technique (ART), has been successfully developed by the author from first principles (i.e., a direct implementation of the mathematics, as opposed to a rewrite of someone else's code) using the Interactive Data Language (IDL). The detailed information about IDL is given in section 2.4. For ground based tomography calculations the program accepts the GPS slant TEC. Mostly the GPS TEC is verified by comparison with TOPEX TEC measurements before being supplied to tomographic algorithm. The tomographic reconstruction of electron density structure has been successfully validated by comparing the results with more direct measurements of ionospheric structure by ground based (such as ionosonde density profiles) and other satellite techniques (such as density profiles extracted from LEO satellites, e.g. CHAMP, by using the occultation method). The tomographic technique has been then applied to study the response of the ionosphere to magnetic storms and atmospheric gravity waves over the mid-latitude Australian region. The tomographic inversion technique that the author has developed has been applied to large amounts of data obtained on different days (between September 1999 and August 2003), especially during magnetically disturbed periods.

This thesis contains seven chapters. This chapter, Chapter 1, gives an overview of the near Earth environment and its behaviour, and especially the phenomena that influence the electron density distribution of the ionosphere. A brief history of the discovery of the ionosphere is also given. Chapter 2 is devoted to the description of the experimental instruments, data, and the software packages that have been used. Here the background history of both GPS and TOPEX satellites, and their data format are briefly described. The GPS TEC reduction principles are described in Chapter 3 in more detail. A brief review of earlier ionospheric structure studies using the transionospheric radio propagation experiment are outlined in Chapter 4. It also includes details of TOPEX and GPS TEC comparison techniques. Most interesting observations of the response of the ionosphere to magnetic storm and atmospheric gravity waves over the Australian region, which extends from equatorial regions to high latitude regions, are also presented in Chapter 4. Chapter 5 is dedicated to describing the tomographic inversion techniques in more detail. The validation of the software is also presented in Chapter 5. The most interesting tomographic results obtained during the campaign that relate to ionospheric irregularities that occur during magnetically disturbed conditions are given in Chapter 6. During the course of this research work, the tomographic inversion technique has also been applied to FedSat's (Australian first ever LEO satellite in 30 years) GPS data to image the ionosphere above the orbiting height (~830 km), which is the region of the plasmasphere, and the most fascinating results obtained are presented in Chapter 6. Finally, Chapter 7 summarises the entire work presented and points out possible future improvement that could be achieved in the resolution of reconstructed electron densities obtained by tomography.